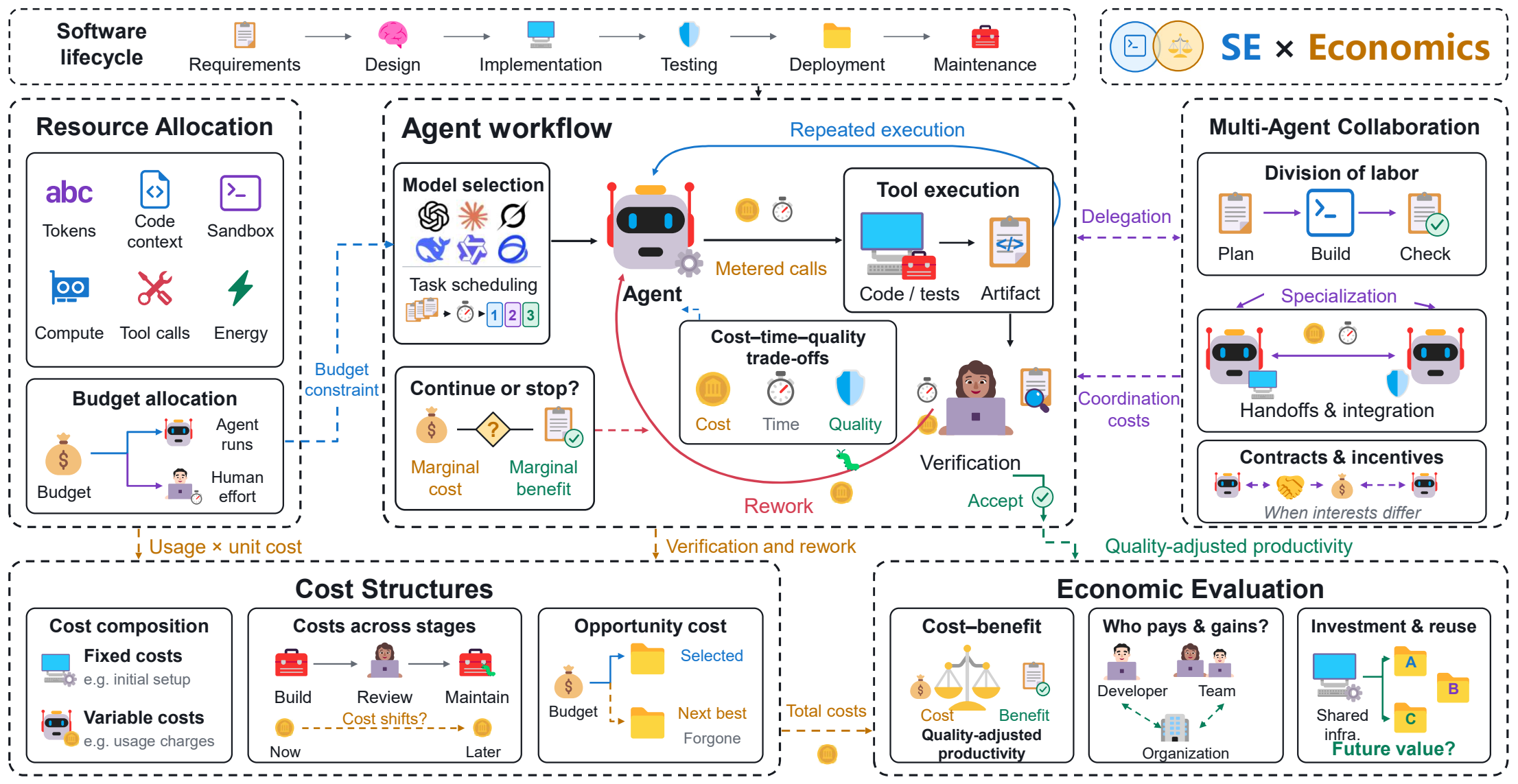

Software lifecycle
Requirements
Design
Implementation
Testing
Deployment
Maintenance
SE × Economics
Resource Allocation
Tokens
Code context
Sandbox
Compute
Tool calls
Energy
Budget allocation
Budget
Agent runs
Human effort
Budget constraint
Agent workflow
Repeated execution
Model selection
Task scheduling
Agent
Metered calls
Tool execution
Code / tests
Artifact
Cost–time–quality trade-offs
Cost
Time
Quality
Continue or stop?
Marginal cost
Marginal benefit
Rework
Verification
Accept
Delegation
Coordination costs
Multi-Agent Collaboration
Division of labor
Plan
Build
Check
Specialization
Handoffs & integration
Contracts & incentives
When interests differ
Usage × unit cost
Verification and rework
Quality-adjusted productivity
Cost Structures
Cost composition
Fixed costs
e.g. initial setup
Variable costs
e.g. usage charges
Costs across stages
Build
Review
Maintain
Cost shifts?
Now
Later
Opportunity cost
Budget
Selected
Next best
Forgone
Total costs
Economic Evaluation
Cost–benefit
Cost
Benefit
Quality-adjusted productivity
Who pays & gains?
Developer
Team
Organization
Investment & reuse
Shared infra.
A
B
C
Future value?